



University of Science and Technology, Chittagong (USTC)
Department of Computer Science and Engineering

LAB REPORT

Course Title: Computer Graphics and Image Processing Lab

Lab Course Code: CSE 418

Submitted to:

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Semester: 7th

Experiment No: 01

Experiment Name: Implementation of Image to Image Translation using Python.

Objective

To understand basic geometric transformations in digital image processing and implement image translation.

Implementation

```
UB_Translation1.ipynb ☆ ☁
File Edit View Insert Runtime Tools Help
Q Commands + Code + Text ▶ Run all

[1] import cv2
import numpy as np
import matplotlib.pyplot as plt

[3] image = cv2.imread('/content/mango_back.jpg')
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

[4] #image dimention
h, w = image.shape[:2] #height, width, channelq

[14] tx = 10
ty = 20

[15] #translation matrix
M= np.float32([
    [1,0,tx],
    [0,1,ty]
])

[15] #translation matrix
M= np.float32([
    [1,0,tx],
    [0,1,ty]
])

[16] translated = cv2.warpAffine(image,M, (w,h))

[17] plt.subplot(1,2,1)
plt.imshow(image)
plt.title('Original')
plt.axis('off')

plt.subplot(1,2,2)
plt.imshow(translated)
plt.title('Translated')
plt.axis('off')

plt.show()
```

Output

Original



shutterstock.com · 2625659447

Translated



Experiment No: 02

Experiment Name: Implementation of Image to Image Scaling using Python.

Objective

To understand basic geometric transformations in digital image processing and implement image scaling.

Implementation

```
UB_ImageScaling.ipynb
File Edit View Insert Runtime Tools Help

Q Commands + Code + Text ▶ Run all

[18]
✓ Os
import cv2
import numpy as np
import matplotlib.pyplot as plt

[19]
✓ Os
image = cv2.imread('/content/mango_back.jpg')
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

[20]
✓ Os
#image dimation
h, w = image.shape[:2] #height, width, channel

[21]
✓ Os
#scaling factors
scale_x = 2 # scale width by 1.5x
scale_y = 1.5 # scale height by 1.5x

[22]
✓ Os
#computing new dimation
new_w = int(w * scale_x)
new_h = int(h * scale_y)

[23]
✓ Os
scaled = cv2.resize(image, (new_w, new_h), interpolation=cv2.INTER_LINEAR)

UB_ImageScaling.ipynb
File Edit View Insert Runtime Tools Help

Q Commands + Code + Text ▶ Run all

[24]
✓ Os
plt.subplot(1, 2, 1)
plt.imshow(image)
plt.title('Original')
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(scaled)
plt.title('Scaled')
plt.axis('off')

plt.show()
```

Output

...

Original



shutterstock.com - 2625659447

Scaled



shutterstock.com - 2625659447

Experiment No: 03

Experiment Name: Implementation of Image to Image Mirroring using Python.

Objective

To understand basic geometric transformations in digital image processing and implement horizontal and vertical image mirroring.

Implementation

```
UB_ImageMirror.ipynb ☆ Saving...
File Edit View Insert Runtime Tools Help

Q Commands + Code + Text ▶ Run all

[1] ✓ 0s
import cv2
import numpy as np
import matplotlib.pyplot as plt

[2] ✓ 0s
image = cv2.imread('/content/pandaUB.sample.jpg')
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

[3] ✓ 0s
#image dimention
h, w = image.shape[:2] #height, width, channel

[4] ✓ 0s
# 1 = horizontal flip (mirror left-right)
# 0 = vertical flip (mirror top-bottom)
# -1 = both (horizontal + vertical)
Mirrored = cv2.flip(image, 1)
```

```
UB_ImageMirror.ipynb ☆
File Edit View Insert Runtime Tools Help

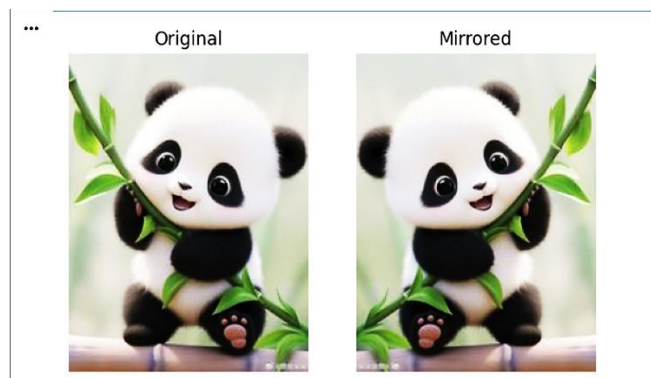
Q Commands + Code + Text ▶ Run all

[5] ✓ 1s
plt.subplot(1, 2, 1)
plt.imshow(image)
plt.title('Original')
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(Mirrored)
plt.title('Mirrored')
plt.axis('off')

plt.show()
```

Output



Experiment No: 04

Experiment Name: Implementation of Image to Image Rotation using Python.

Objective

To understand basic geometric transformations in digital image processing and implement image rotation.

Implementation

```
UB_imageRotation.ipynb
File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[8] import cv2
import numpy as np
import matplotlib.pyplot as plt

[9] image = cv2.imread('/content/pandaUB.sample.jpg')
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

[10] #image dimention
h, w = image.shape[:2] #height, width, channel

[11] #Rotation matrix
Center = (w / 2, h / 2)
Angle = 45 # degrees, positive
Scale = 1.0 # to keep original size

M = cv2.getRotationMatrix2D(Center, Angle, Scale)

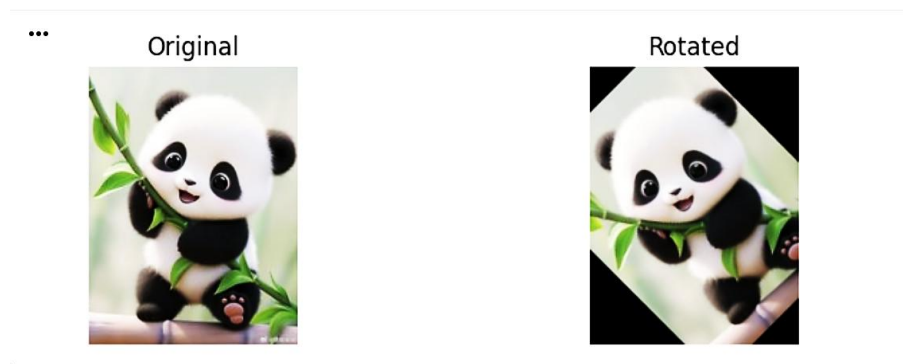
[12] Rotated = cv2.warpAffine(image, M, (w, h))

[13] plt.subplot(1, 3, 1)
plt.imshow(image)
plt.title('Original')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(Rotated)
plt.title('Rotated')
plt.axis('off')

plt.show()
```

Output



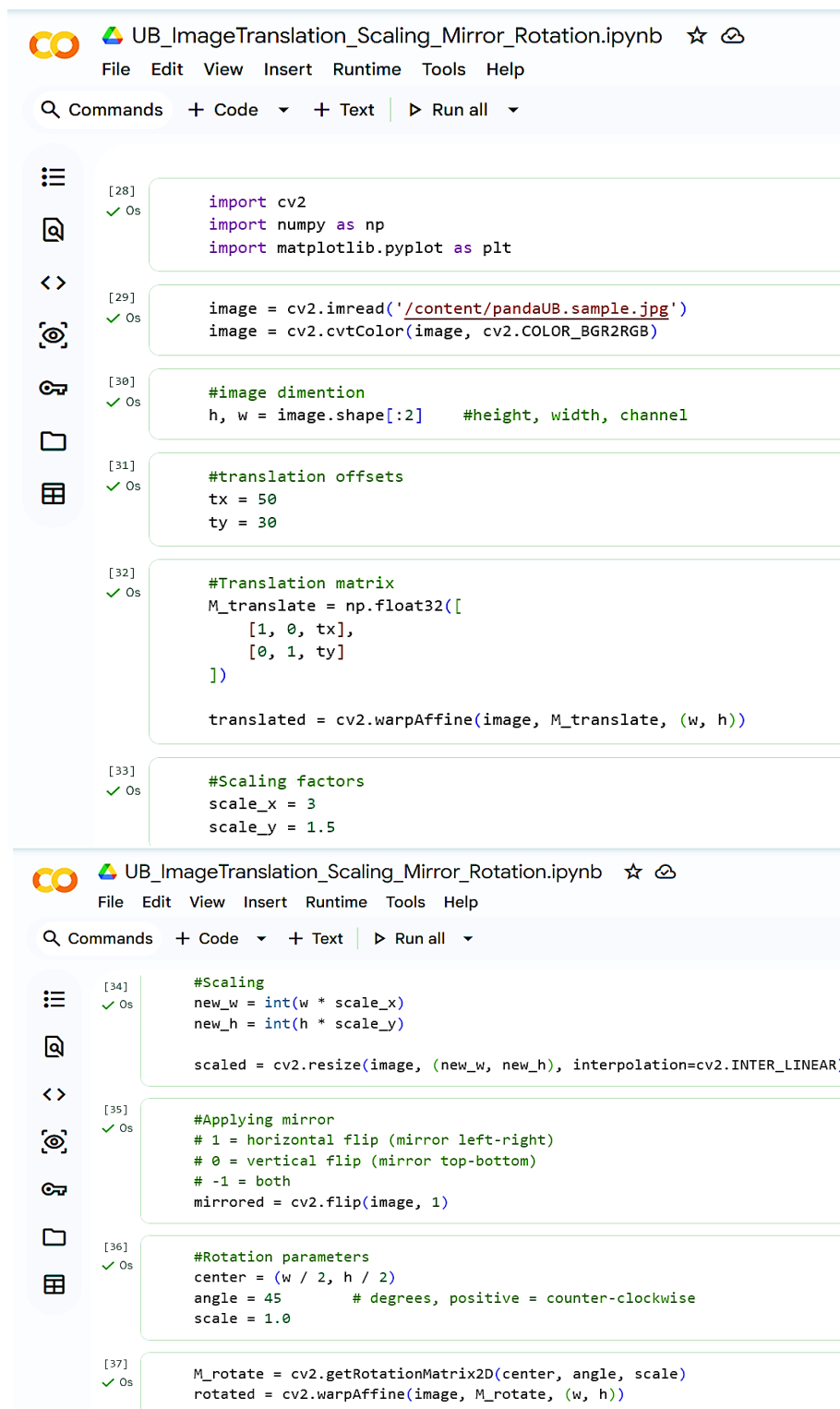
Experiment No: 05

Experiment Name: Implementation of Image to Image Transformations Using Python.

Objective

To combine multiple transformations (image translation, scaling, mirroring, rotation) in a single program and observe the visual effects of different transformations on an image.

Implementation



```
UB_ImageTranslation_Scaling_Mirror_Rotation.ipynb ☆ ☁
File Edit View Insert Runtime Tools Help
Q Commands + Code + Text ▶ Run all

[28] ✓ 0s
import cv2
import numpy as np
import matplotlib.pyplot as plt

[29] ✓ 0s
image = cv2.imread('/content/pandaUB.sample.jpg')
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

[30] ✓ 0s
#image dimention
h, w = image.shape[:2] #height, width, channel

[31] ✓ 0s
#translation offsets
tx = 50
ty = 30

[32] ✓ 0s
#Translation matrix
M_translate = np.float32([
    [1, 0, tx],
    [0, 1, ty]
])

translated = cv2.warpAffine(image, M_translate, (w, h))

[33] ✓ 0s
#Scaling factors
scale_x = 3
scale_y = 1.5

UB_ImageTranslation_Scaling_Mirror_Rotation.ipynb ☆ ☁
File Edit View Insert Runtime Tools Help
Q Commands + Code + Text ▶ Run all

[34] ✓ 0s
#Scaling
new_w = int(w * scale_x)
new_h = int(h * scale_y)

scaled = cv2.resize(image, (new_w, new_h), interpolation=cv2.INTER_LINEAR)

[35] ✓ 0s
#Applying mirror
# 1 = horizontal flip (mirror left-right)
# 0 = vertical flip (mirror top-bottom)
# -1 = both
mirrored = cv2.flip(image, 1)

[36] ✓ 0s
#Rotation parameters
center = (w / 2, h / 2)
angle = 45 # degrees, positive = counter-clockwise
scale = 1.0

[37] ✓ 0s
M_rotate = cv2.getRotationMatrix2D(center, angle, scale)
rotated = cv2.warpAffine(image, M_rotate, (w, h))
```

Q Commands + Code + Text ▶ Run all ▼

```
[38] ✓ 4s ▶ plt.figure(figsize=(18, 10))

plt.subplot(2, 3, 1)
plt.imshow(image)
plt.title('Original')
plt.axis('off')

plt.subplot(2, 3, 2)
plt.imshow(translated)
plt.title('Translated')
plt.axis('off')

plt.subplot(2, 3, 3)
plt.imshow(scaled)
plt.title('Scaled')
plt.axis('off')

plt.subplot(2, 3, 4)
plt.imshow(mirrored)
plt.title('Mirrored')
plt.axis('off')

plt.subplot(2, 3, 5)
plt.imshow(rotated)
plt.title('Rotated')
plt.axis('off')

plt.tight_layout()
plt.show()
```

Output

